

# Somjit Nath

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## Experience

Nov 2019 – present **Researcher**, *Planning & Control*, Tata Consultancy Services Research & Innovation, Mumbai, India.  
*Supervisor: Dr. Harshad Khadilkar*

## Education

2017 – 2019 **Master of Science (Thesis)**, *University of Alberta*, Edmonton, Canada.  
Computing Science.  
CGPA: 4.00/4.00  
*Supervisor: Dr. Martha White*

2013 – 2017 **Bachelor of Engineering**, *Jadavpur University*, Kolkata, India.  
Electrical Engineering.  
*First Class Honours, CGPA: 9.08/10.00, 85.52%*

2013 **Higher Secondary Examination**, *South Point High School*, Kolkata, India.  
Science (Mathematics, Physics, Chemistry); Elective - Statistics.  
*First Division, 88.4%*

2011 **Madhyamik**, *South Point High School*, Kolkata, India.  
Elective - Mechanics.  
*First Division, 90%*

## Publications

Aug 2021 **Revisiting State Augmentation methods for Reinforcement Learning with Stochastic Delays**, *Conference on Information and Knowledge Management (CIKM) 2021*, Somjit Nath, Mayank Baranwal and Harshad Khadilkar.

May 2021 **Scalable Multi-Product Inventory Control with Lead Time Constraints using Reinforcement Learning**, *Neural Computing and Applications*, Hardik Meisheri, Nazneen N Sultana, Mayank Baranwal, Vinita Baniwal, Somjit Nath, Satyam Verma, Balaraman Ravindran, Harshad Khadilkar.

- May 2021 **SIBRE: Self Improvement Based REwards for Adaptive Feedback in Reinforcement Learning**, *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021, Somjit Nath, Richa Verma, Abhik Ray, Harshad Khadilkar.
- April 2020 **Training Recurrent Neural Networks Online by Learning Explicit State Variables**, *International Conference on Learning Representations (ICLR)*, 2020, Somjit Nath, Vincent Liu, Alan Chan, Xin Li, Adam White, Martha White.
- May 2019 **Two-Timescale Networks for Nonlinear Value Function Approximation**, *International Conference on Learning Representations (ICLR)*, 2019, Wesley Chung, Somjit Nath, Ajin Joseph, Martha White.
- Dec 2018 **A Fixed-Point Formulation for Recurrent Neural Networks**, *Continual Learning Workshop, Neural Information Processing Systems(NeurlIPS)*, 2018, Somjit Nath, Taher Jafferjee and Martha White.
- Dec 2018 **Rejection Sampling for Off-Policy Learning**, *Continual Learning Workshop, Neural Information Processing Systems(NeurlIPS)*, 2018, Wesley Chung, Sina Ghiassian, Somjit Nath and Martha White.
- April 2018 **Smartphone Camera Based Analysis of ELISA using Artificial Neural Network**, *IET Computer Vision*, Somjit Nath, Subhannita Sarcar, Biswendu Chatterjee, Nabendu Chatterjee, Rhishita Chourashi.
- Dec 2016 **Arduino Based Door Unlocking System with Real Time Control**, *IEEE International Conference on Contemporary Computing and Informatics (IC3I)*, Somjit Nath, Paramita Banerjee, Rathindra Nath Biswas, Swarup Kumar Mitra and Mrinal Kanti Naskar.

## Teaching Experience

- Jan - April 2018 **Graduate Teaching Assistant**, *Department of Computing Science, University of Alberta, Edmonton, Canada.*  
CMPUT 275, Introduction to Tangible Computing-II
- Sept - Dec 2017 **Graduate Teaching Assistant**, *Department of Computing Science, University of Alberta, Edmonton, Canada.*  
CMPUT 274, Introduction to Tangible Computing-I

## Internships

- June 2016 **Vocational Trainee**, *Steel Authority of India IISCO*, Burnpur, India.
- May – Jul 2015 **Research Intern**, *Indian Statistical Institute*, Kolkata, India.  
Computer Vision and Pattern Recognition Department

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## Scholastic Achievements

- May 2017 **Scholarship for Academic Excellence**, *State Electrical Engineers' Association*, Kolkata, India.
- Feb 2016 **2nd in Autonomous Robotics event "Sherlock"**, *KSHITIJ, the technology fair (largest in Asia) held at Indian Institute of Technology(IIT), Kharagpur*, India.
- 2015 **Summer Fellow**, *Indian Academy of Sciences*, Bengaluru, India.
- Feb 2014 **2nd in Autonomous Robotics event**, *SRIJAN, the technology fair held at Jadavpur University*, Kolkata, India.

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## Technical skills

- Language C,C++,Python,HTML,PHP,Javascript,SQL,VHDL
- Tools Tensorflow, PyTorch, MATLAB, Octave, OpenCV, Arduino, Processing
- Operating System Windows, UNIX, MacOSX

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## Extra-curricular Activities

- 2018 Love to play Table Tennis and participated in tournaments – Edmonton Chinatown Open 2018 & CUSTTA Open 2018, Calgary
- Mar 2016 Member of the organizing committee of the hackathon "SPARKHACK" and the circuit-building competition "CIRCUISTIC" held at the annual department technology fair "CONVOLUTION '16", Department of Electrical Engineering, Jadavpur University, Kolkata, India
- Mar 2016 Mentored a group of five school students for a robotics competition in "CONVOLUTION' 16", Department of Electrical Engineering, Jadavpur University, Kolkata, India